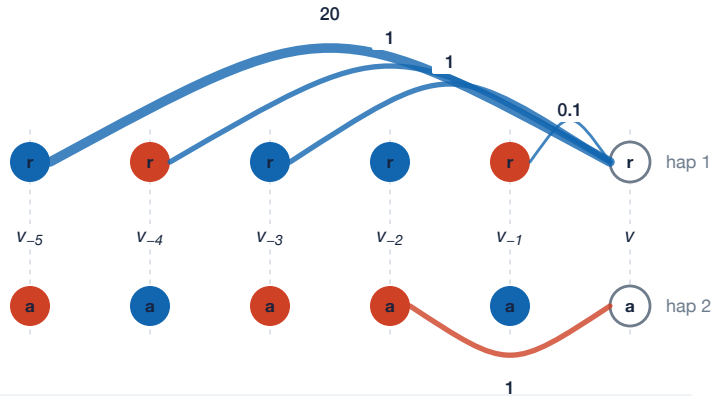


a Weighted votes from up to k = 35 upstream variants



$$h_1(v) = 20 + 1 + 1 + 0.1 = 22.1$$

$$h_2(v) = 1$$

$v \rightarrow$ haplotype 1 (a tie opens a new phase block)

c Block formation



b Phasing entropy

$$PE(v) = - \sum p_i \log_2 p_i, p_i = h_i / (h_1 + h_2)$$

22.1 : 1 PE = 0.257

3 : 1 PE = 0.811 PE ≥ 0.80

4 : 3 PE = 0.985

5 : 5 PE = 1.000

$PE \geq 0.80 \rightarrow$ a GNN window is opened (Suppl. Fig. 6)

written to VCF as INFO/PE, INFO/H1, INFO/H2

allele on haplotype 1

allele on haplotype 2